

Calibrating One-Round Membership Inference with Neighbors

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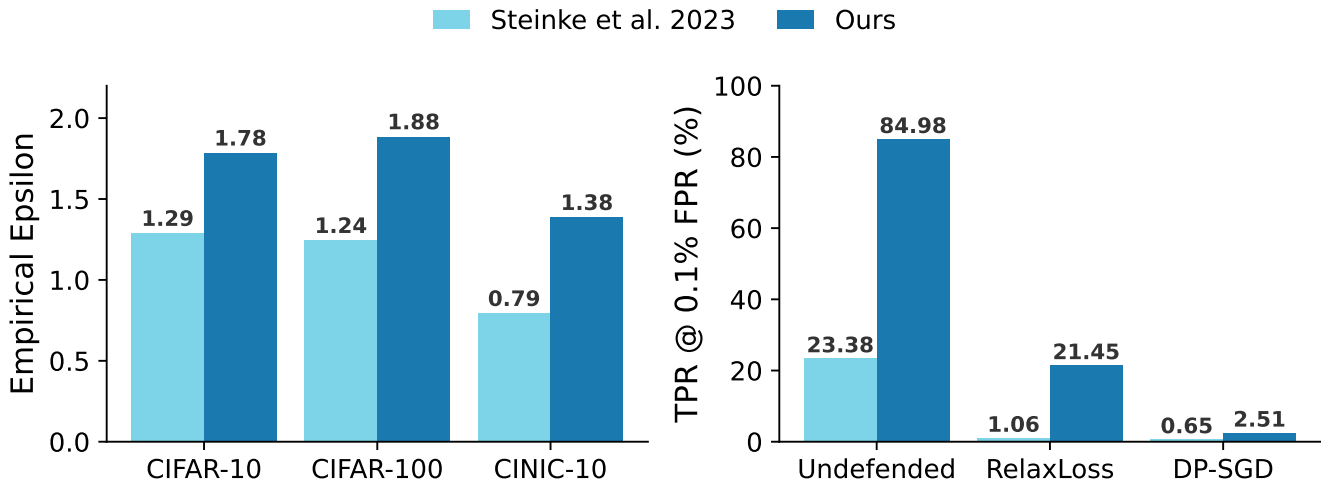


Figure 1: Comparison between Steinke et al. [25] and our best method. On the left we report the empirical ϵ in the DP-SGD setting across CIFAR-10, CIFAR-100 and CINIC-10. On the right we report the TPR at 0.1% FPR on CIFAR-10 across three privacy settings: undefended models, RelaxLoss and DP-SGD.

Abstract

The state-of-the-art Membership Inference (MI) methods calibrate their signal separately for each example using reference models, auxiliary models trained to exclude the target. This paradigm scales poorly to modern large models, however, whose training is too expensive to replicate. This has motivated one-round settings, where only a single trained model is available; but without reference models the per-example calibration that drives the strongest attacks can no longer be estimated, leaving the membership signal weak. We ask whether neighbors of the target point can recover this calibration without training any additional model. Our key observation is that reference models serve only to reveal how an example behaves under models not trained on it, and that querying the target model on nearby samples yields the same information. We propose two complementary ways to obtain such neighbors, and show that querying them against an early training checkpoint further sharpens the signal. We evaluate across three image classification datasets and three training setups, showing that neighbors yield strong membership signals and competitive attack performance at no additional training cost.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **General and reference** → *Evaluation*; • **Security and privacy**;

Keywords

machine learning; privacy; audit; membership inference; DP-SGD

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1 Introduction

Membership inference attacks [24] have become a standard tool for auditing the privacy of machine learning models. Given a trained model and a target sample, an MIA decides whether that sample was part of the model’s training set. Because models tend to behave differently on data they have memorized than on unseen data, the success of such an attack offers an empirical, sample-level measure of how much a model leaks about its training data.

The strongest MIAs achieve this by *calibrating* the target’s behavior on a per-example basis. Rather than asking whether the loss on a sample is low in absolute terms, they ask whether it is surprisingly low *for that particular sample*, accounting for the fact that some examples are intrinsically easier than others [4]. This calibration is precisely what separates state-of-the-art attacks from naive loss



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thresholding, and it is obtained through *reference* (or *shadow*) models: auxiliary models that reveal how the target model would score a sample when that sample is excluded from training [2, 4, 11, 24].

This paradigm, however, has grown increasingly impractical with the advent of highly scalable architectures such as Transformers [27] and the large foundation models they enabled [9, 26]. Training even a single such model can be prohibitively expensive; training the tens or hundreds of reference models that calibration requires is often out of reach. This has motivated growing interest in the *one-round* setting, where the auditor has access to a single trained model and cannot retrain new models. The one-round regime is especially central to the auditing of differentially private (DP) training, where one seeks an empirical lower bound on the privacy parameter ϵ from a single training run, precisely to avoid the cost of repeated retraining [17, 25, 29].

Removing reference models, however, removes the very ingredient that makes modern attacks strong. Without them, the per-example calibration can no longer be estimated, and existing one-round methods fall back on a global loss threshold or simple heuristics [25]. The resulting signal is weak, particularly at the low false-positive rates that matter most for membership inference [4], which in turn loosens DP audits. Recovering per-example calibration *without* reference models is therefore the common bottleneck for both one-round attacks and one-run DP auditing.

In this work, we show that this calibration can be recovered using nothing beyond the target model itself. Our starting point is the observation that reference models serve a single purpose: to reveal how the target sample would score under models that did not train on it. We obtain the same information differently, by querying the target model on samples that *neighbor* the target point. Intuitively, the model’s behavior on a non-member sample close to the target approximates the behavior a reference model would exhibit on the target itself. Figure 2 contrasts the two views: where reference-based attacks query the target sample against many models, we query many neighbors against a single model. Building on this idea, we characterize what makes a neighbor useful and propose two complementary ways to construct neighbors, by retrieving them from held-out data (*Similarity Neighbors*) and by synthesizing them with a diffusion model (*Synthetic Neighbors*); in the context of internal audits, we further show that querying neighbors against an *early training checkpoint* sharpens the signal further. Across three datasets and three privacy regimes, our attacks substantially outperform the one-round baseline of Steinke et al. [25] and close much of the gap to reference-based attacks, without training a single additional model, as previewed in Figure 1.

2 Background and Preliminary

2.1 Membership Inference: Setup and Threat Model

Let F denote a model trained on a dataset D drawn from a data distribution $\mathbb{D}$. *Membership inference* (MI) asks, for a given example x , whether $x \in D$. Following the standard formulation [24, 30], an MI adversary $\mathcal{A}$ outputs a score reflecting its belief that x was a training member; thresholding this score yields a membership decision, and the score’s quality is measured by the resulting true-positive/false-positive trade-off. Within this framework, we refer to

canaries as the subset of training examples that are most vulnerable to membership inference.

Threat model. We study the most constrained and most realistic regime, which we refer to as *one-round* membership inference. The adversary is given a single target model F_t and an example x , and may query F_t to obtain a per-example signal such as the loss or confidence on x . Crucially, the adversary *cannot* train additional models. This rules out the reference-model machinery on which the strongest attacks rely, and it is the setting faced by an auditor or adversary confronting a single deployed model.

2.2 From Loss Thresholding to Reference-Based Attacks

The simplest attack thresholds the per-example loss: if $L(F_t, x)$ is small, predict “member” [30]. This is weak because examples differ in intrinsic difficulty. Some examples have low loss whether or not they were trained on, while others have high loss regardless, so a single global threshold conflates easy non-members with true members. The membership signal comes not from the magnitude of the loss but from *calibrating* that loss to the specific example.

Reference-model (or *shadow-model*) attacks supply this calibration [4, 24]. By training auxiliary models on data drawn from $\mathbb{D}$, the adversary estimates how a model behaves on x when x is, or is not, in its training set, and calibrates the target’s behavior against this reference. The Likelihood Ratio Attack (LiRA) [4] makes this precise: it casts membership as a per-example hypothesis test, comparing the target’s score on x against the distribution of scores that models *not* trained on x assign to x .

2.3 Offline LiRA, Formally

Let $s(F, x)$ be a membership score for example x under model F . A standard choice is the logit-scaled confidence [4]

$$s(F, x) = \log \frac{p}{1-p}, \quad p = F(x)_y, \quad (1)$$

where $F(x)_y$ is the predicted probability on the true label y ; larger s indicates stronger memorization of x by F . The logit scaling is used because s is approximately Gaussian, whereas the raw confidence is not.

Offline LiRA assumes access only to *OUT* models—models not trained on x —and performs a one-sided test.¹ Concretely, it trains a set of *OUT* shadow models $\{F_1, \dots, F_n\}$, none of which is trained on x , and approximates the *OUT* score distribution as a Gaussian

$$\mathcal{D}_1 = \mathcal{N}(\mu_{\text{out}}, \sigma_{\text{out}}^2), \quad (2)$$

with

$$\mu_{\text{out}} = \frac{1}{n} \sum_{i=1}^n s(F_i, x), \quad \sigma_{\text{out}}^2 = \frac{1}{n} \sum_{i=1}^n (s(F_i, x) - \mu_{\text{out}})^2. \quad (3)$$

Membership is then assessed by how surprising the target’s score is under $\mathcal{D}_1$:

$$\Lambda(x) = 1 - \Phi\left(\frac{s(F_t, x) - \mu_{\text{out}}}{\sigma_{\text{out}}}\right), \quad (4)$$

¹This is the appropriate baseline for our setting: the *online* variant additionally trains *IN* models that include x , which requires retraining with the target example and is therefore unavailable in the one-round regime. For brevity, we henceforth write *LiRA* to denote its offline variant unless stated otherwise.

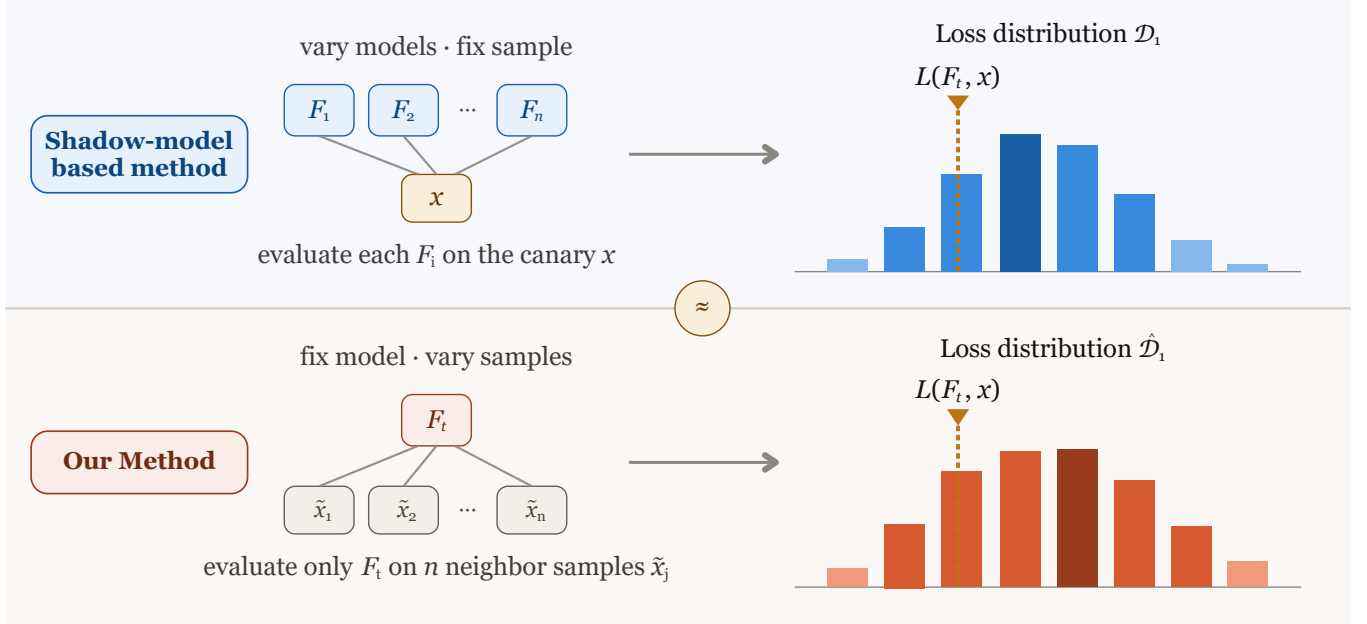


Figure 2: Comparison between shadow-model based methods and our method. Instead of characterizing the loss distribution by querying the canary against different shadow models, our method queries different samples neighboring the canary against the target model alone.

where Φ is the standard normal CDF. A small $\Lambda(x)$ —the target’s score lying in the upper tail of $\mathcal{D}_1$ —indicates that x is a member.

The entire procedure reduces to one object: the per-example OUT distribution $\mathcal{D}_1$, estimated from a set of OUT scores $\{s(F_i, x)\}_{i=1}^n$. Its strength comes entirely from this per-example calibration, and its cost comes entirely from obtaining the n OUT models needed to form it.

2.4 One-Round MI and One-Run DP Auditing

The reference models underlying Eq. (3) are precisely what the one-round setting denies. Without the ability to train OUT models, $\mathcal{D}_1$ cannot be estimated per example, and existing one-round methods fall back on a global loss threshold or simple per-sample heuristics—discarding the calibration that drives the signal. The result is a faint membership signal, especially in the low-false-positive-rate regime that matters most for MI [4].

The same constraint motivates the auditing of differentially private (DP) training [12]. DP auditing derives an empirical lower bound on the privacy parameter ϵ from the distinguishability of members and non-members, and has moved decisively toward the *single-run* regime [25]: rather than retraining a model many times, one inserts many canary examples into a single training run and infers ϵ from their membership scores, precisely to avoid the prohibitive cost of repeated training. The tightness of such an audit—how close the empirical ϵ comes to the true privacy loss—is governed by the separation between member and non-member scores. When each canary is scored by its global loss, this separation is poor, and the audit is correspondingly loose, forcing either an underestimate of ϵ or an impractically large number of canaries.

Recovering per-example calibration *without* OUT models is therefore the common bottleneck for both one-round attacks and one-run auditing. In Section 3, we address it directly: we estimate the OUT distribution $\mathcal{D}_1$ using the target model’s scores on *neighbors* of x , requiring nothing beyond the target model itself.

3 Calibrating One-Round MI with Neighbors

LiRA owes its strength to a single object: the per-example OUT distribution $\mathcal{D}_1$ of Eq. (2), whose mean and variance are estimated from the scores that OUT shadow models assign to the example under test—the *canary*, in the auditing setting. As noted in Section 2.3, the shadow models serve no purpose beyond producing this set of scores. We exploit this observation directly: rather than training shadow models, we form the same set of scores by querying the target model F_t on samples that *neighbor* the canary (see Figure 2). The intuition, which we make precise below, is that the target’s behavior on nearby non-member samples proxies the canary’s behavior across OUT models.

Formally, our attack is a single substitution in the LiRA estimator. Given a set of n neighbors $\{\tilde{x}_1, \dots, \tilde{x}_n\}$ of the canary x , we replace the OUT-model scores $\{s(F_i, x)\}$ with the target’s scores on the neighbors and estimate

$$\hat{\mu}_{\text{out}} = \frac{1}{n} \sum_{j=1}^n s(F_t, \tilde{x}_j), \quad \hat{\sigma}_{\text{out}}^2 = \frac{1}{n} \sum_{j=1}^n (s(F_t, \tilde{x}_j) - \hat{\mu}_{\text{out}})^2, \quad (5)$$

yielding $\hat{\mathcal{D}}_1 = \mathcal{N}(\hat{\mu}_{\text{out}}, \hat{\sigma}_{\text{out}}^2)$. We then score the canary by the tail probability of $s(F_t, x)$ under $\hat{\mathcal{D}}_1$, exactly as in LiRA (Eq. (4)). The full procedure is given in Algorithm 1; it requires no model training and queries only F_t , making it applicable in the one-round setting.

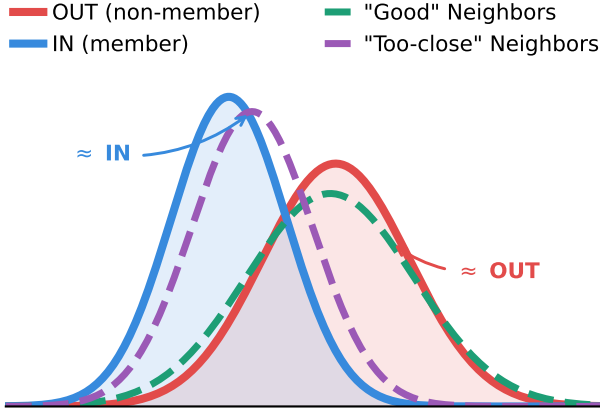


Figure 3: Comparison of shadow model-based and neighbor-driven confidence distributions for a member canary. The IN (blue) and OUT (red) distributions are estimated using shadow models trained with and without the canary, respectively. Neighbor distributions are obtained by querying Stable Diffusion img2img generations against the target model. Neighbors that are too similar (strength 0.45) resemble the member (IN) distribution, whereas stronger perturbations (strength 0.75) better approximate the non-member (OUT) distribution.

3.1 What Makes a “Good” Neighbor

The validity of the substitution in Eq. (5) rests on one assumption, which we state explicitly.

ASSUMPTION 1 (LOCAL SMOOTHNESS). *The target model’s membership score $s(F_t, \cdot)$ varies smoothly over the data manifold in a neighborhood of the canary x . Consequently, for a non-member $\tilde{x}$ sufficiently close to x , the score $s(F_t, \tilde{x})$ approximates the score an OUT model assigns to x , i.e. a sample from $\mathcal{D}_1$.*

Assumption 1 exposes a distance trade-off intrinsic to our estimator. To faithfully approximate $\mathcal{D}_1$, each neighbor must be close enough to the canary to be representative of its behavior. Yet being too close is harmful: if the canary was seen during training, the target memorizes it, and a neighbor that is too similar has its score pulled toward the member regime, contaminating $\hat{\mathcal{D}}_1$ and shrinking the very membership signal we seek. For member canaries the trade-off depends further on intrinsic learning difficulty and the model’s memorization capacity: a hard-to-memorize sample exhibits similar scores regardless of membership, tolerating closer neighbors, whereas an easy-to-memorize sample has a wide membership-dependent score gap, requiring neighbors to lie further away.

Because membership labels are unavailable at inference time, this trade-off must be respected whether or not the canary is a member. Even if one could condition on membership, the optimal distance would still depend on canary- and model-specific characteristics that vary considerably across samples and settings. These factors make it difficult to formulate a single distance threshold whose tuning reliably optimizes the trade-off across all canaries. We therefore do not attempt to pin down one optimal distance. Instead,

Algorithm 1 Neighbor approximation of LiRA. In the Early Checkpoint variant, the neighbor scores in line 2 are computed against an early checkpoint of the target model.

Require: target model F_t , canary x , number of neighbors n

```

1:  $\{\tilde{x}_1, \dots, \tilde{x}_n\} \leftarrow \text{neighbors}(x)$     {search / generate neighbors}
2:  $S_{\text{out}} \leftarrow \{s(F_t, \tilde{x}_1), \dots, s(F_t, \tilde{x}_n)\}$ 
3:  $\hat{\mu}_{\text{out}} \leftarrow \text{mean}(S_{\text{out}})$ 
4:  $\hat{\sigma}_{\text{out}}^2 \leftarrow \text{var}(S_{\text{out}})$ 
5: return  $1 - \text{GaussianTail}(s(F_t, x); \hat{\mu}_{\text{out}}, \hat{\sigma}_{\text{out}}^2)$ 
```

we keep neighbors within a controlled similarity *band*—close to the canary, yet not too close—and operationalize this band through two complementary construction mechanisms: the intrinsic train–test gap when retrieving neighbors from the test set (Section 3.2), and the denoising strength when synthesizing them (Section 3.3). Both mechanisms use a notion of distance to *locate* candidates near the canary, but neither relies on a single tuned distance threshold as its decision criterion.

3.2 Similarity Neighbors

The test set is a natural source of neighbors: by construction it consists entirely of non-member data, shares the training distribution, and is held at the intrinsic train–test distance from any training sample—precisely the upper edge of the band described in Section 3.1. Exploiting this, we select each canary’s neighbors by retrieving the most similar test samples.

Concretely, given a canary we first discard all test samples not in its class. We then embed the canary and each remaining test sample with OpenAI’s CLIP ViT-B/32 model [20] and compute their cosine similarity. To obtain a more robust score, we repeat this over 18 augmentations of the input pair (horizontal shifts, vertical shifts, and vertical flips) and average the resulting scores. Simpler variants that omit the class filter or the augmentations are possible; we adopt this formulation as it yields the best empirical performance. As shown in Section 4.5, it already outperforms Steinke et al. [25] in most scenarios.

3.3 Synthetic Neighbors

Our second construction shifts from existing data to synthetic neighbors. This offers greater flexibility, as the generated samples can be controlled through the choice of generative model, its hyperparameters, and optional text prompting—and it applies even when the test set contains no suitable neighbors. We generate neighbors with the image-to-image pipeline of Stable Diffusion² [21], which produces variations of an input image by partially noising and then denoising it.

The pipeline is governed by three core hyperparameters:

- **Number of steps.** The total number of denoising iterations.
- **Strength.** A value in $[0, 1]$ controlling the diffusion step from which denoising begins, effectively determining how much the output may deviate from the input.
- **Guidance scale.** The influence of the text prompt on generation.

²We use the stable-diffusion-v1-5 implementation, as the original RunwayML model is deprecated.

Strength is the mechanism by which we control the similarity band of Section 3.1: to enforce sufficient distance between canary and neighbors, we consider only medium-to-high strength values, granting higher variance from the original image. To assess the potential of this strategy, we report the highest score across *number of steps* $\in \{30, 35, 40\}$, *strength* $\in \{0.55, 0.65, 0.75, 0.85\}$, and *guidance scale* $\in \{1, 3, 5\}$. For guidance scale 1 we use an empty prompt; for larger values we use a short class-representative prompt (e.g., “A dog”, “An airplane”). As shown in Section 4.5, this approach consistently outperforms both the baseline Steinke et al. [25] and our first proposal.

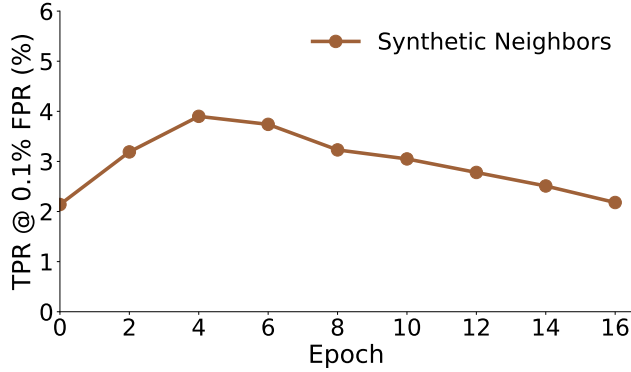


Figure 4: TPR at 0.1% FPR of our attack when querying synthetic neighbors against early training checkpoints, from a real example.

3.4 Early Checkpoints

To ease the distance trade-off of Section 3.1, we propose evaluating the neighbors’ scores against an *early checkpoint* of the target model rather than the fully trained one. We call this the Early Checkpoint (E.C.) variant. Unlike our base attacks, this variant relaxes the black-box assumption, as it requires access to intermediate training snapshots rather than the fully trained model alone; it is therefore best suited to internal auditing settings.

The intuition is that before memorization sets in, the model has acquired generalization but has not yet overfit specific training examples. At this stage the score landscape around a member canary resembles that of non-member samples, so even closer neighbors serve as good non-member representatives. Figure 4 corroborates this intuition: the attack TPR peaks in the early epochs and progressively declines as training continues.

Since we cannot know a priori when memorization begins, we periodically snapshot the model during training and select the checkpoint yielding the highest privacy leakage. Crucially, the early checkpoint is used *only* to compute the neighbor scores that characterize $\hat{\mathcal{D}}_1$; the canary score is still evaluated against the fully trained model. In Algorithm 1 this amounts to replacing F_t with an early checkpoint exclusively in line 2.

We note that this variant relaxes the black-box assumption: it requires access to intermediate training states. This is essentially free in the one-run *auditing* setting, where the auditor controls

training and can save checkpoints at no cost, but it is a stronger assumption for an external *attacker*, for whom it applies only when such checkpoints are available. In Section 4.5 we show that, where applicable, E.C. consistently improves performance across all our methods.

4 Experiments

Following Aerni et al. [1], we adopt their model architectures and training pipelines, in accordance with their recommendation to evaluate privacy defenses on high-utility models. We evaluate each attack across three datasets and three privacy settings, resulting in nine experimental scenarios. In every scenario, the objective is to infer the membership of 500 canaries across 64 independently trained models.

Canaries are randomly selected from the training set and assigned independently to half of the models. Each model is trained on all non-canary samples together with its assigned canaries. This assignment ensures that every canary appears equally often as a member and a non-member sample across the ensemble, enabling a balanced evaluation of membership inference performance. To increase memorization and maximize privacy leakage, the labels of all canaries are flipped prior to training.

4.1 Datasets, Defenses and Metrics

We conduct experiments on CIFAR-10, CIFAR-100, and CINIC-10 [8], three widely used image-classification benchmarks in the privacy literature. To increase the distributional difference between CIFAR-10 and CINIC-10, we remove all CIFAR-10 images from CINIC-10, as their presence would otherwise introduce overlap between the two datasets and reduce the diversity of our evaluation. We then subsample the remaining data to 50k training and 10k test examples, matching the scale of CIFAR-10 while preserving class balance.

Prior auditing work primarily focuses on estimating the differential privacy parameter ϵ and therefore evaluates attacks exclusively against DP-SGD [25]. Our objective, however, is not to obtain the tightest privacy guarantees, but rather to assess the quality of membership scores that can be computed without auxiliary models. Under this perspective, privacy leakage serves as a proxy for score quality rather than an end goal in itself. Consequently, the evaluation need not be restricted to ϵ estimation or to differentially private models. We therefore consider three privacy regimes: undefended training, RelaxLoss [6], and DP-SGD, representing no protection, empirical leakage mitigation, and formal privacy guarantees, respectively.

To evaluate privacy leakage, we report TPR at 0.1% FPR in all settings and, for DP-SGD models, the empirical ϵ . Following Carlini et al. [4], we adopt TPR at low FPR because low false-positive rates are critical in realistic membership inference scenarios and provide a direct measure of attack effectiveness. For DP-SGD, we additionally estimate ϵ using the procedure of Steinke et al. [25], evaluating all pairs $k_+, k_- \in \{10, 20, \dots, 250\}$, retaining the tightest estimate for each model, and reporting the average across the 64-model ensemble.

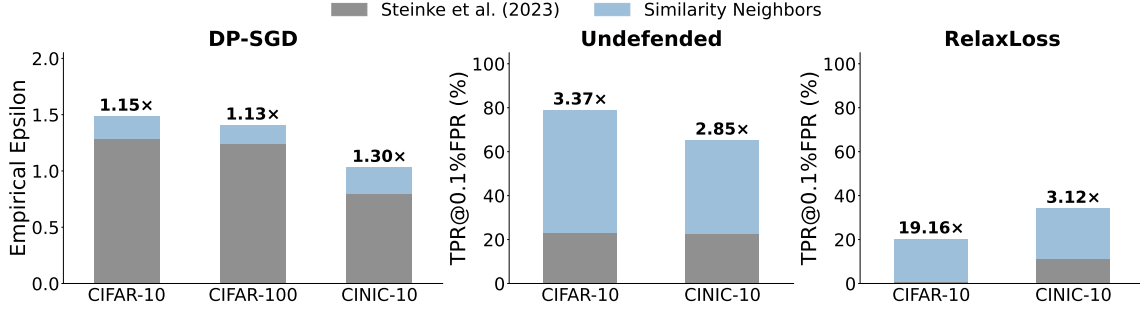


Figure 5: Settings where our Similarity Neighbors approach significantly outperforms Steinke et al. [25].

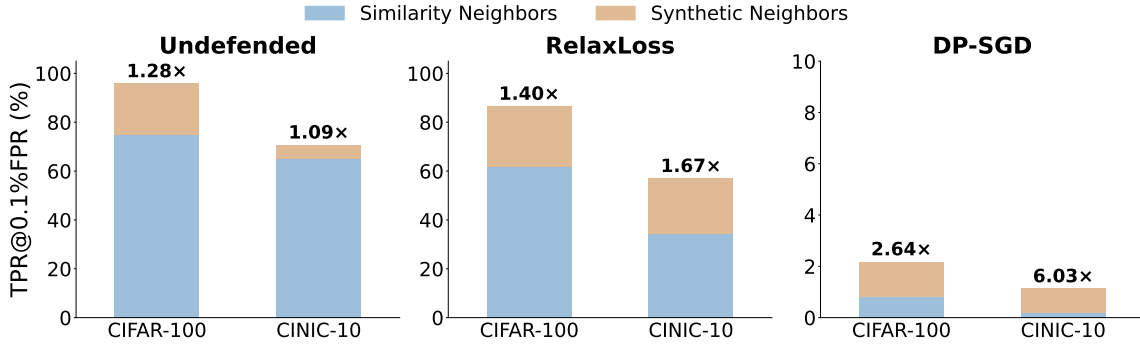


Figure 6: Settings where our Synthetic Neighbors approach significantly outperforms our Similarity Neighbors approach.

Since TPR at low FPR considers only the highest-scored members, while ϵ estimation aggregates over both members and non-members, the two metrics can diverge. Intuitively, ϵ captures average-case leakage: it reflects how well the attack discriminates overall across the score distribution. TPR at low FPR, by contrast, is a worst-case metric: it measures whether the attack can confidently identify *any* members, regardless of its behavior on the rest. To see why these can disagree, consider two attacks with identical ϵ estimates: one concentrates its correct predictions at the top of the ranking (high TPR at low FPR), the other at the bottom (near-zero TPR at low FPR). This asymmetry motivates reporting both metrics jointly in DP settings.

4.2 Similarity Neighbors

In Figure 5, we present representative settings where our Similarity Neighbors approach significantly outperforms Steinke et al. [25]. Looking at both metrics, the results follow a consistent trend. Under DP-SGD, our attack achieves modest but meaningful improvements over the baseline in terms of empirical ϵ across all three datasets, suggesting that our scores are consistently stronger in the traditional one-round DP auditing setting. The TPR results reveal a particularly pronounced effect. Against undefended models, our attack substantially outperforms the baseline across both CIFAR-10 and CINIC-10, indicating that our method is far more effective at identifying members when no empirical defense is active. The gap becomes even more dramatic under RelaxLoss, where our attack achieves almost 20 $\times$ higher TPR on CIFAR-10, demonstrating that

our approach is considerably more robust to this defense than the baseline.

Overall, these results highlight the importance of per-sample calibrated scores. They further confirm that strong calibrated signals can be effectively achieved within a one-round setting through neighbor-based approximations relying solely on test data—without the need for shadow model training or auxiliary data collection.

4.3 Synthetic Neighbors

Figure 6 reports a selection of settings in which Synthetic Neighbors outperforms Similarity Neighbors, focusing on the cases where the performance gap is most pronounced. Against undefended models, Synthetic Neighbors already yields a substantial advantage on CIFAR-100, reaching nearly 96% TPR, while the gap narrows on CINIC-10. Under RelaxLoss, Similarity Neighbors suffer a marked degradation across both datasets, whereas Synthetic Neighbors maintain considerably higher TPR, indicating greater resilience to this form of defense. The contrast is most striking under DP-SGD: despite both methods being heavily suppressed by the strong privacy guarantees, Synthetic Neighbors achieve disproportionately higher TPR across both datasets, pointing to a qualitative difference in attack effectiveness.

These results highlight a consistent and defense-agnostic advantage of Synthetic Neighbors, which we attribute to the flexibility of the generative approach. Unlike Similarity Neighbors, whose effectiveness is inherently limited by the availability of similar samples

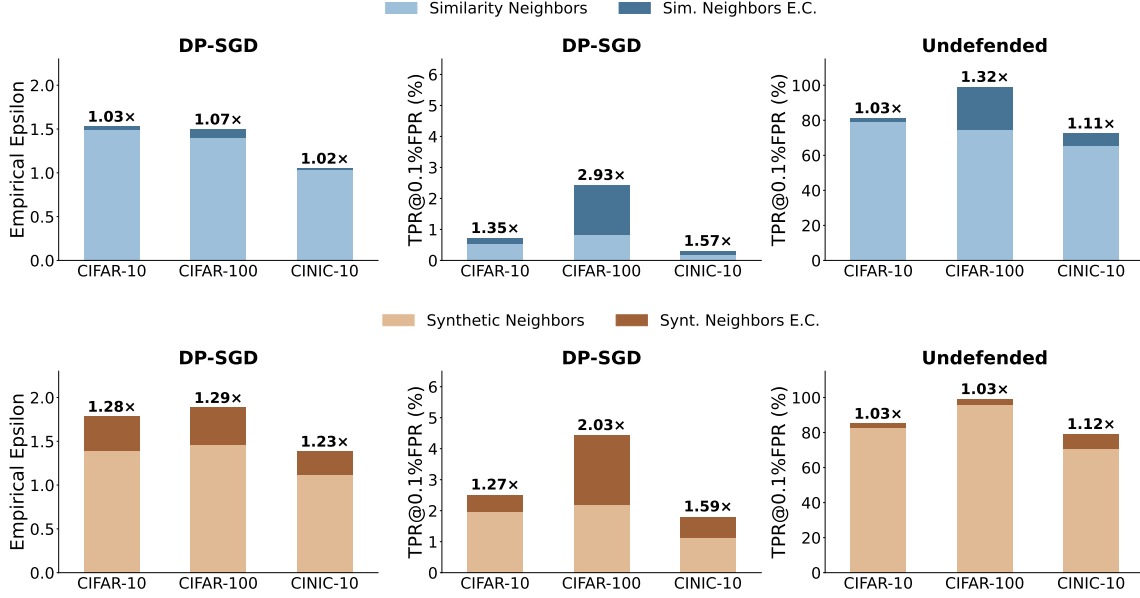


Figure 7: Settings where our Early Checkpoint (E.C.) variants significantly outperform the standard attacks.

in the test set, Synthetic Neighbors can produce informative neighbors for virtually any canary, making the attack robust to cases where the test distribution fails to provide neighbors of adequate quality.

4.4 Early Checkpoints

The results reported in Figure 7 consistently demonstrate that the Early Checkpoint (E.C.) variants outperform their standard counterparts across both DP-SGD and undefended settings, both metrics, and all three datasets.

For Similarity Neighbors, empirical epsilon improvements under DP-SGD are modest, with the E.C. variant providing only marginal gains across all datasets. The picture changes more clearly when looking at TPR at 0.1% FPR, where the E.C. variant achieves notably stronger results, particularly on CIFAR-100. In the undefended setting, the improvement is most dramatic on CIFAR-100, where the standard variant reaches 74.9% while the E.C. variant nearly saturates at 99.0%, with more moderate but still consistent gains on CIFAR-10 and CINIC-10.

For Synthetic Neighbors, the E.C. variant delivers more substantial empirical epsilon improvements than in the similarity-based case, with consistent gains across all three datasets. TPR improvements under DP-SGD are consistent across all datasets, with CIFAR-100 again showing the largest relative gain. In the undefended setting, the E.C. variant approaches saturation on CIFAR-100 and delivers meaningful gains on CINIC-10, while CIFAR-10 shows a smaller margin due to the standard variant already performing strongly there.

Taken together, these results make a compelling case for the E.C. variants. The gains are not isolated to a specific attack family, dataset, or defense regime — they are systematic. The effect is especially pronounced in the most challenging settings, precisely

where standard attacks struggle most. This robustness across diverse experimental conditions strongly supports querying neighbors against early checkpoints as a broadly effective strategy, consistently yielding stronger membership inference signals regardless of the underlying neighbor generation method.

4.5 Main Results

Considering the results reported in the Sections 4.2, 4.3 and 4.4, Synthetic Neighbors E.C. emerges as our strongest proposal. In Tables 1 and 2, we therefore expand the direct comparison with Steinke et al. [25], already anticipated in Figure 1. We further report results for LiRA, as upper bound on our method.

In Table 1, we measure the privacy leakage against DP-SGD in terms of ϵ , showing that our attack yields improvements over the baseline ranging from roughly 0.5 to 0.6 across the datasets, while consistently retaining more than 50% of the LiRA performance.

In Table 2, we measure the TPR at 0.1% FPR across all nine settings. Against undefended models, we exhibit only marginal performance loss compared to LiRA, outperforming Steinke et al. [25] by a factor of $\sim 3.5\times$ on both CIFAR-10 and CINIC-10. On CIFAR-100, the higher task complexity limits the maximum achievable test accuracy, leading to a more pronounced train-test accuracy gap, which makes overfitting particularly pronounced and drives nearly all attacks to saturate near 100% TPR. Against DP-SGD, despite the restricted privacy leakage, our approach still outperforms the baseline by a factor of approximately $4\times$ on CIFAR-10, $2\times$ on CIFAR-100, and $5\times$ on CINIC-10. Against RelaxLoss, our method again substantially outperforms the baseline across all datasets, by a factor of $20\times$ on CIFAR-10 and $5\times$ on CINIC-10, while achieving an intermediate TPR between it and LiRA on CIFAR-100.

Overall, Synthetic Neighbors E.C. consistently and significantly closes the gap to the LiRA upper bound across all privacy settings

Table 1: Comparison between Steinke et al. [25] (baseline), our Synthetic Neighbors E.C. attack and LiRA (upper bound) against DP-SGD, in terms of empirical ϵ .

Empirical Epsilon		Steinke et al. [25]	Ours	LiRA
DP-SGD	CIFAR-10	1.29	1.78	2.99
	CIFAR-100	1.24	1.88	3.16
	CINIC-10	0.79	1.38	2.37

Table 2: Comparison between Steinke et al. [25] (baseline), our Synthetic Neighbors E.C. attack and LiRA (upper bound) across all privacy settings, in terms of TPR at 0.1% FPR.

TPR@0.1%FPR (%)		Steinke et al. [25]	Ours	LiRA
Undefended	CIFAR-10	23.38	84.98	97.03
	CIFAR-100	99.28	99.03	99.73
	CINIC-10	22.88	79.10	91.29
DP-SGD	CIFAR-10	0.65	2.51	12.21
	CIFAR-100	2.19	4.43	13.89
	CINIC-10	0.33	1.80	4.43
RelaxLoss	CIFAR-10	1.06	21.45	69.24
	CIFAR-100	72.68	84.64	98.31
	CINIC-10	10.96	53.73	78.46

and datasets, substantially outperforming Steinke et al. [25] in every setting. These results highlight that more powerful and calibrated MI scores directly translate into tighter empirical lower bounds on ϵ in the context of one-round DP auditing, and, more broadly, into a more accurate estimation of the true privacy leakage of a model.

5 Related Work

5.1 Reference-Based Membership Inference Attacks

As anticipated in Section 2.2, Membership inference attacks (MIAs) have traditionally relied on shadow models—also referred to as reference models—as a mechanism for approximating the behavior of a target model. The shadow-model paradigm was introduced by Shokri et al. [24], who proposed training multiple auxiliary models on datasets drawn from the same distribution as the target model and subsequently using their outputs to train an attack classifier that distinguishes members from non-members. This work established the canonical black-box MIA framework and demonstrated that membership information can be inferred from model predictions alone.

Subsequent work sought to improve both the efficiency and fidelity of this paradigm. Salem et al. [22] showed that effective membership inference can be achieved with significantly fewer shadow models and even without a dedicated attack classifier, suggesting that the core membership signal lies in the statistical differences between member and non-member predictions rather than in the complexity of the attack model itself. More recently, Li et al. [15] proposed ℓ -Leaks, arguing that posterior probabilities discard valuable information and showing that shadow models trained

to characterize the distribution of pre-softmax logits can substantially improve attack performance. These developments reflect a broader shift from heuristic attack-classifier pipelines toward statistically grounded methods that explicitly model the distributions induced by training membership. This evolution culminated in the Likelihood Ratio Attack (LiRA) [4], which formulates membership inference as a hypothesis-testing problem. By explicitly modeling uncertainty and focusing on the low-false-positive-rate regime, LiRA consistently outperforms previous shadow-model-based attacks and is widely regarded as the state-of-the-art reference-model attack, making it the primary benchmark against which modern membership inference methods are evaluated.

Despite providing stronger membership signals, *Online* LiRA comes at a higher computational cost than *Offline* LiRA, as it relies also on *IN* models—models trained on the target sample. Building on the notation introduced in Section 2.3, alongside the set of *OUT* shadow models $\{F_1, \dots, F_n\}$, *Online* LiRA trains a set of *IN* shadow models $\{F'_1, \dots, F'_n\}$ to approximate the *IN* score distribution as a Gaussian

$$\mathcal{D}_0 = \mathcal{N}(\mu_{\text{in}}, \sigma_{\text{in}}^2), \quad (6)$$

with

$$\mu_{\text{in}} = \frac{1}{n} \sum_{i=1}^n s(F'_i, x), \quad \sigma_{\text{in}}^2 = \frac{1}{n} \sum_{i=1}^n (s(F'_i, x) - \mu_{\text{in}})^2. \quad (7)$$

Membership is then assessed by the target’s score likelihood ratio between $\mathcal{D}_0$ and $\mathcal{D}_1$:

$$\Lambda(x) = \frac{\mathcal{N}(s(F_t, x); \mu_{\text{in}}, \sigma_{\text{in}}^2)}{\mathcal{N}(s(F_t, x); \mu_{\text{out}}, \sigma_{\text{out}}^2)} \quad (8)$$

5.2 Reference-Free Membership Inference Attacks

The high computational cost of shadow model training has motivated a parallel line of work seeking membership signals that require no reference models. The simplest such approaches are metric-based, exploiting statistics directly computable from the target model. Yeom et al. [30] proposed thresholding on the model’s loss, exploiting the observation that members tend to incur lower loss than non-members. Bertran et al. [3] proposed an alternative route via quantile regression, training a lightweight auxiliary model to directly estimate the target model’s score distribution without any shadow training, reporting competitive performance against LiRA at low false-positive rates.

For language models specifically, Carlini et al. [5] refined loss-based inference by normalizing perplexity against zlib entropy, partially correcting for the intrinsic complexity of each sample, while token-level methods such as Min-K% [23] and Min-K%++ [31] aggregate statistics over the lowest-probability tokens, grounded in the intuition that non-members are more likely to contain outlier tokens. Beyond general-purpose attacks, shadow-free methods have also been proposed for specific settings, including recommender systems [7] and federated learning [10].

5.3 One-Round DP Audits

Although reference-free attacks reduce the per-attack cost, reliably estimating the privacy leakage—such as TPR at low FPR—still requires evaluating a large number of samples across multiple models.

One-round auditing addresses this orthogonal bottleneck in DP settings by shifting the auditing target to a formal ϵ lower bound, constraining the access to a single trained target model. While averaging over multiple runs yields more representative estimates, a statistically valid ϵ bound can in principle be obtained from a single training run.

Steinke et al. [25] first introduced ORA by auditing randomly included canary points into a single training run, and converting membership guesses into an ϵ lower bound via a binomial tail bound under (ϵ, δ) -DP. Mahloujifar et al. [17] tightened this by grounding the conversion in the full f -DP trade-off curve and Xiang et al. [29] further derived the theoretically optimal membership decoder under f -DP via order statistics over per-canary privacy loss scores.

Despite these advances largely contributing to refine the statistical procedure that converts membership guesses to privacy lower bounds, the membership signal itself has been largely taken for granted—typically model loss or gradient projections. On the theoretical side, Keinan et al. [14] show that interference between canary signals is the dominant barrier to tight auditing in DP-SGD, directly motivating the design of signals with higher per-canary distinguishability. Notably, Liu et al. [16] substitute the loss-based signal with a quantile regression-based membership inference attack, building on previous work from Bertran et al. [3]. Rather than relying on a global loss threshold, their method trains an auxiliary model to predict sample-specific score quantiles, yielding per-example membership decisions that better account for heterogeneity across canaries and reporting substantial gains in black-box settings. However, this still requires training an auxiliary model, albeit one that is independent of the target and can be kept arbitrarily small. In contrast, our method requires no additional model training whatsoever, bridging the gap between the statistical power of reference-based attacks and the computational efficiency demanded by one-run auditing.

5.4 Neighbors and synthetic data in Membership Inference Attacks

The use of neighboring or synthetic samples is not new in the MIA literature. Early works leveraged generative models to compensate for the lack of access to the target model’s training distribution. For instance, Zhang et al. [32] enrich the available data with synthetic samples, enabling the training of shadow models even when only limited information about the original training set is available. Beyond data augmentation, synthetic samples can also be used to probe the local behavior of a model around a target point. Wen et al. [28] argue that a target sample alone provides limited information for membership inference. They therefore employ adversarial tools to optimize the generation of diverse perturbations around a canary sample, allowing for a more informative exploration of its surrounding region and improving inference performance.

This intuition has recently gained particular attention also in the context of MIAs against large language models (LLMs), where neighboring samples are used as reference points to characterize model behavior. Mattern et al. [18] observe that reference-model-based attacks rely on the often unrealistic assumption of access to samples drawn from the training distribution. To overcome this limitation, they compare the model’s score on the target sample

with scores obtained on synthetically generated neighboring texts, eliminating the need for reference data. Similarly, Galli et al. [13] exploit neighboring samples by querying *noisy* variants of the target instance, obtained through stochastic perturbations in the embedding space at inference time. Finally, Mozaffari and Marathe [19] extend this idea by generating semantically perturbed versions of the target text and training a neural classifier to distinguish members from non-members based on the model’s responses across these perturbations.

6 Limitations

As discussed in Section 3, the core difficulty underlying our approach is that the optimal distance between a canary and its neighbors depends on canary- and model-specific factors that are not known a priori, and that cannot be resolved without membership labels. We are unable to formulate a metric that reliably quantifies neighbor quality ahead of time; instead, we rely on constructions (similarity-based retrieval, diffusion strength, early checkpoints) that implicitly control this trade-off. Developing a principled, canary-specific criterion for neighbor quality remains an open problem, and we see it as the most promising direction to further close the gap to reference-based attacks.

Beyond this core limitation, each of our proposed constructions carries its own constraints. Since Similarity Neighbors are retrieved from the test set, the attack’s effectiveness for a given canary is inherently limited by whether the test set contains samples that lie in an adequate similarity band around it; when no such neighbors exist, the estimated $\hat{D}_1$ is a poor approximation and the attack underperforms. Synthetic Neighbors, in turn, require a generative model capable of producing suitable variations of the input, which may not be readily available or effective for all data modalities or tasks, and depend on hyperparameters whose optimal range must be explored per setting; this exploration can be more or less costly depending on how thoroughly it is performed. As noted in Section 3.4, the Early Checkpoint variant further relaxes the black-box assumption, as it requires access to intermediate training checkpoints that are not generally available to an external attacker; it is therefore best suited to internal auditing settings, where the auditor controls training and checkpoints come at no additional cost, rather than to external attacks.

Finally, the scale of our models and experiments is limited, for two reasons. First, prior work on one-round DP auditing, including the baseline [25], evaluates on datasets and models of similar scale, so our choice keeps our results directly comparable to existing literature. Second, our evaluation protocol requires training 64 independent models across nine experimental scenarios to obtain reliable estimates of attack performance; replicating this protocol at the scale of large foundation models would demand computational resources far beyond what is feasible in this work. We leave the evaluation of our method on larger-scale models, including LLMs to future work.

7 Conclusion

In this work, we introduce two novel classes of black-box membership inference attacks that require no additional model training, making them suitable for one-round audits. While prior work on

one-round auditing focuses on refining the procedure that estimates the privacy leakage from membership scores, we address a complementary and largely unexplored direction: the design of powerful membership scores that require no auxiliary models. We propose to approximate the out score distribution of Offline LiRA by querying neighbors of the canary on the target model alone, and instantiate this idea in two ways: the Similarity Neighbors attack, which retrieves neighbors from the test set using a CLIP-based similarity metric, and the Synthetic Neighbors attack, which generates neighbors via the image-to-image pipeline of Stable Diffusion. We further propose an Early Checkpoint variant of both attacks, which evaluates neighbors on intermediate snapshots of the target model to mitigate the effect of memorization on the IN canaries.

Our experimental evaluation across three datasets and three privacy defenses demonstrates that both proposals consistently outperform Steinke et al. [25]. The Synthetic Neighbors attack and its Early Checkpoint variant emerge as the strongest attacks in virtually every setting, achieving particularly large TPR gains in the DP-SGD setting and maintaining strong performance under Undefended and RelaxLoss scenarios. The Similarity Neighbors attack, on the other hand, provides a simpler yet powerful alternative that requires no additional data generation, making it an attractive option when synthetic generation is not feasible.

More broadly, our results suggest that meaningful approximations of state-of-the-art MIAs are achievable in one-round settings. We hope this work encourages further investigation into model-free membership inference attacks, and contributes to a more comprehensive understanding of one-round audits beyond the DP-SGD setting.

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The source code and scripts required to reproduce our experiments are available on GitHub at `ethz-spylab/one_round_MI`. Instructions about how to get started and the structure of the repository are contained in the `README.md` file.

In Tables 3 and 4 we report all our experimental results.

B Use of Generative AI

As stated in Section 3.3, we used the `stable-diffusion-v1-5` generative model to perform our attack relying on synthetic neighbors. Moreover, the writing of this paper has been assisted by LLMs.

DP-SGD	Empirical Epsilon			TPR@0.1%FPR (%)		
	CIFAR-10	CIFAR-100	CINIC-10	CIFAR-10	CIFAR-100	CINIC-10
Steinke et al. [25]	1.2867	1.2427	0.7917	0.6500	2.1875	0.3312
Similarity Neighbors	1.4851	1.4062	1.0305	0.5250	0.8250	0.1875
Sim. Neighbors E.C.	1.5294	1.4981	1.0504	0.7063	2.4188	0.2938
Synthetic Neighbors	1.3926	1.4549	1.1216	1.9688	2.1812	1.1312
Synt. Neighbors E.C.	1.7832	1.8837	1.3849	2.5063	4.4250	1.8000
LiRA	2.9934	3.1560	2.3675	12.2125	13.8875	4.4250

Table 3: Experimental results against DP-SGD, across CIFAR-10, CIFAR-100 and CINIC-10. We measure both ϵ and TPR at 0.1% FPR. Comparison between Steinke et al. [25] (baseline), our Similarity and Synthetic Neighbors attacks with their Early Checkpoint (E.C.) variants and LiRA (upper bound).

TPR@0.1%FPR (%)	Undefended			RelaxLoss		
	CIFAR-10	CIFAR-100	CINIC-10	CIFAR-10	CIFAR-100	CINIC-10
Steinke et al. [25]	23.3813	99.2812	22.8813	1.0563	72.6750	10.9625
Similarity Neighbors	78.9062	74.8875	65.2125	20.2437	61.9687	34.1500
Sim. Neighbors E.C.	81.1063	99.0062	72.4125	20.9663	82.0187	34.4750
Synthetic Neighbors	82.6375	95.8625	70.7562	19.9875	86.6188	57.0500
Synt. Neighbors E.C.	84.9813	99.0250	79.1000	21.4500	84.6361	53.7312
LiRA	97.0250	99.7313	91.2937	69.2375	98.3062	78.4563

Table 4: Experimental results against undefended models and RelaxLoss, across CIFAR-10, CIFAR-100 and CINIC-10. We evaluate the TPR at 0.1% FPR. Comparison between Steinke et al. [25] (baseline), our Similarity and Synthetic Neighbors attacks with their Early Checkpoint (E.C.) variants and LiRA (upper bound).